

Concept for Further Test Enhancements

1. Additional Checks

I propose the following additional checks to enhance our testing process:

- **Performance Testing:** I will implement tests to measure the loading time of the onboarding process and document upload. This will help identify any performance bottlenecks that may affect user experience.
- **Security Testing:** I plan to conduct tests to check for vulnerabilities such as SQL injection, XSS, and CSRF during the registration and document upload processes to ensure the application is secure.
- **Accessibility Testing:** I will ensure that the application is accessible to users with disabilities by using tools like Axe or Lighthouse to automate accessibility checks.
- **Cross-Browser Testing:** I intend to validate that the application works across different browsers (Chrome, Firefox, Safari) and devices (desktop, mobile) to ensure a consistent user experience.
- **Error Handling:** I will test how the application handles errors during registration and document upload (e.g., invalid email format, file size limits) to improve robustness.

2. Recommended New Tests

I recommend implementing the following new tests:

- **End-to-End Tests for Edge Cases:** I will create tests for edge cases, such as registering with an already used email, uploading unsupported file types, and submitting the form with empty required fields.
- **User Journey Tests:** I plan to implement tests that simulate a complete user journey, from registration to document upload and posting a question, to ensure a seamless experience.
- **API Testing:** If the application has a backend API, I will consider implementing tests to validate API responses for registration and document upload.
- **Mobile-Specific Tests:** I will create tests that specifically check the mobile responsiveness and functionality of the onboarding process.

3. Non-Automatable Properties

I have identified several aspects of the onboarding and document upload processes that exhibit non-automatable properties:

- **Human Interaction:** Some aspects of the onboarding process may require human judgment (e.g., user experience, design feedback). I will consider using manual testing for these areas.
- **Dynamic Content:** If the application displays dynamic content that changes frequently (e.g., user-generated content), it may be challenging to automate tests for these scenarios. I will use visual regression testing tools to capture UI changes.
- **Third-Party Integrations:** If the onboarding process involves third-party services (e.g., payment gateways, social media logins), these may have non-automatable properties. I will use mocks or stubs for testing.

4. Approach to Testing Non-Automatable Properties

To address the non-automatable properties, I will take the following approach:

- **Manual Testing:** For aspects that cannot be automated, I will establish a manual testing process with clear guidelines and checklists.
- **User Feedback:** I will gather feedback from real users during beta testing to identify areas that require improvement.
- **Visual Testing:** I will use visual testing tools (e.g., Percy, AppliTools) to capture screenshots and compare them against baseline images to identify visual discrepancies.